The Hall–Paige Conjecture

Rainbow Perfect Matchings in Cayley Tables

Yuxuan Lu

Shandong University

October 8, 2026



Contents

1. From Euler to the Hall–Paige Conjecture

2. How Many Transversals? Semiqueens and Queens

3. Beyond Hall–Paige: Subsquares and Cycle Types

Euler's Thirty-Six Officers Problem

A♠	Q♣	J♡	K♢
K♡	J♢	Q♠	A♣
Q♢	A♡	K♣	J♠
J♣	K♠	A♢	Q♡

Figure: A card puzzle from Ozanam (1723): every row and column contains each rank and each suit. It has a solution.

Problem 1 (Euler, 1782)

Six regiments each send six officers, one of each of six ranks. Can the 36 *officers stand in a* 6×6 *square so that every row and every column contains* ***one officer of each regiment*** *and* ***one of each rank****?*

Euler found no arrangement.

Two Latin Squares in One

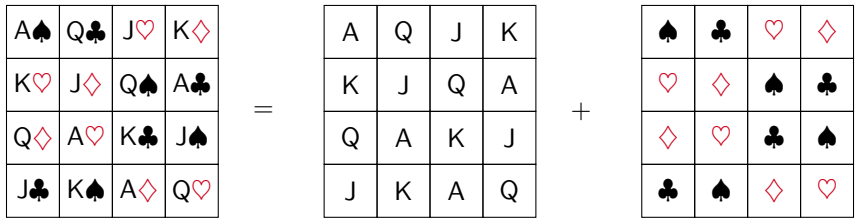


Figure: Forgetting the suits leaves a Latin square of ranks; forgetting the ranks leaves a Latin square of suits. Every rank–suit pair occurs exactly once.

Euler wrote the regiments with Latin letters and the ranks with Greek letters — hence the name **Latin square** (Montgomery, 2024).

Orthogonal Latin Squares and Euler's Conjecture

Definition 2

A **Latin square** of order n is an $n \times n$ array of n symbols in which every symbol occurs once in each row and each column. Two Latin squares $A = (a_{ij})$ and $B = (b_{ij})$ are **orthogonal** if the n^2 pairs (a_{ij}, b_{ij}) are all distinct.

Euler's problem asks for two orthogonal Latin squares of order 6.

Conjecture 1 (Euler, 1782)

For $n \equiv 2 \pmod 4$, there are no two orthogonal Latin squares of order n.

- $\triangleright$ $n = 2$: true, trivially.
- $\triangleright$ (Tarry, 1900) $n = 6$: true — the 36 officers cannot be arranged.
- $\triangleright$ (Bose–Shrikhande, 1959) counterexamples for $n = 22$ and $n = 50$.
- $\triangleright$ (Bose–Shrikhande–Parker, 1960) counterexamples for every $n \equiv 2 \pmod 4$ with $n \geq 10$.

Euler was right only for $n = 2$ and $n = 6$.

Why Statisticians Cared: Latin Squares in Field Trials

▷ R. A. Fisher promoted Latin squares in experiments at Rothamsted (1919–1933) and in *The Design of Experiments* (1935) (Bailey, 2013).

▷ Why rows *and* columns: "on any given field agricultural operations … have followed one of two directions …; consequently streaks of fertility, weed infestation, etc., do, in fact, occur predominantly in those two directions" (Fisher to Jeffreys, 1938, quoted in Bailey).

	Ile de France	Beauce	Champagne	Picardy
20 Feb	potatoes	turnips	beets	oats & peas
20 Mar	turnips	beets	oats & peas	potatoes
20 Apr	beets	oats & peas	potatoes	turnips
20 May	oats & peas	potatoes	turnips	beets

Figure: The oldest known example (Bailey): Cretté de Palluel, 1790, 16 sheep. Rows: slaughter date; columns: breed; entries: diet. Each diet once per date and once per breed.

A Second Square on Top: Graeco-Latin Squares

$A\alpha$	$B\beta$	$C\gamma$
$C\beta$	$A\gamma$	$B\alpha$
$B\gamma$	$C\alpha$	$A\beta$

Figure: Nine officers: each Latin letter meets each Greek letter once (Bailey).

Superimposing a second square **orthogonal** to the first balances a second family of treatments against rows, columns and the first family. Such pairs are called **Graeco-Latin squares** in the statistics literature (Dey, 2017).

Theorem 3 (see Bailey, 2013)

For odd n, the squares $L_1(i,j) = i + j$ and $L_2(i,j) = i - j$ on $\mathbb{Z}_n$ are orthogonal.

Proof. Equal pairs in cells $(i_1, j_1), (i_2, j_2)$ give $i_1 + j_1 = i_2 + j_2$ and $i_1 - j_1 = i_2 - j_2$, so $2(j_1 - j_2) = 0$; as n is odd, $j_1 = j_2$ and $i_1 = i_2$. □

L_1 is the Cayley table of $\mathbb{Z}_n$: odd cyclic groups give orthogonal pairs at once. The problem is even orders.

A Mate Is a Decomposition into Transversals

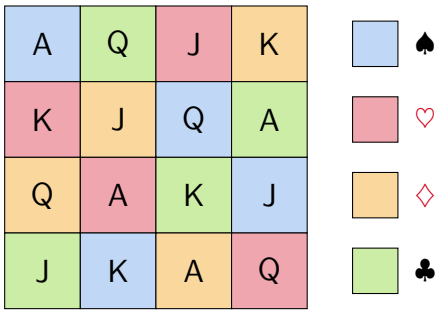


Figure: The ranks, shaded by suit. Each suit picks one cell in every row, every column and every rank.

Definition 4

A **transversal** of a Latin square of order n is a set of n cells, one in each row and each column, with n distinct symbols.

Theorem 5 (Wanless, 2007)

A Latin square has an orthogonal mate $\iff$ *it decomposes into* n *disjoint transversals.*

Euler's question is about one Latin square.

The Abstract Problem: Rainbow Perfect Matchings

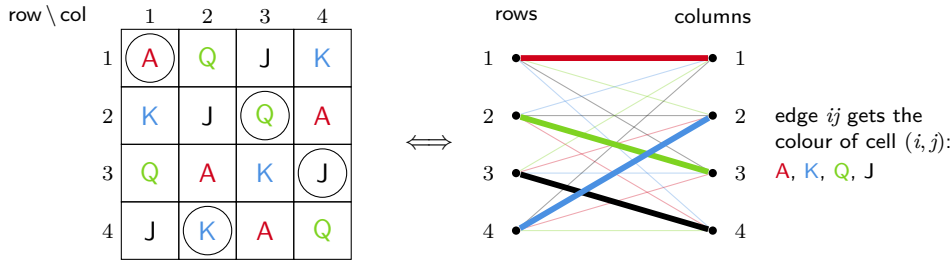


Figure: The ranks as a proper colouring of $K_{4,4}$; the circled transversal is the bold rainbow perfect matching.

Latin square of order n	$\longleftrightarrow$	proper n-edge-colouring of $K_{n,n}$
transversal	$\longleftrightarrow$	**rainbow perfect matching**
orthogonal mate	$\longleftrightarrow$	decomposition into n rainbow perfect matchings

Proposition 1: Mates Are Transversal Decompositions

View a Latin square L of order n as the colouring of $K_{n,n}$ in which the edge (row i, column j) has colour $L(i,j)$; it is **proper** because each symbol occurs once per row and once per column.

Proposition 6 (cf. Wanless, 2007)

*L has an orthogonal mate $\iff$ its cells split into n disjoint transversals $\iff$ $E(K_{n,n})$ splits into n **rainbow perfect matchings**.*

Proof. $\Rightarrow$: for a mate M and a symbol s, the n cells with $M = s$ lie in distinct rows and columns (M is Latin) and carry distinct L-symbols (the pairs (L, M) are distinct): a transversal. Different s give disjoint transversals. $\Leftarrow$: number the transversals $1, \ldots, n$ and let $M(i,j)$ be the number of the one containing (i,j). Each row and column meets each transversal once, so M is Latin; transversal s contains each symbol a once, so (a,s) occurs once. The last equivalence is the dictionary: transversal = rainbow perfect matching. $\square$

Does One Rainbow Perfect Matching Exist?

Euler asked for a **decomposition**. But even **one** rainbow perfect matching need not exist.

Conjecture 2 (Ryser–Brualdi–Stein; Ryser, 1967, Stein, 1975)

Every Latin square of order n has a partial transversal with $n-1$ cells, and a transversal if n is odd.

▷ (Montgomery, 2023) a partial transversal with $n-1$ cells for all large n: the conjecture holds for large **even** n.

▷ The odd case — a full transversal — is open.

The Most Natural Latin Squares: Cayley Tables

Definition 7

The **Cayley table** of a finite group G has rows and columns indexed by G and the entry ab in cell (a, b). It is a Latin square; as a colouring of $K_{n,n}$, the edge ab gets colour ab.

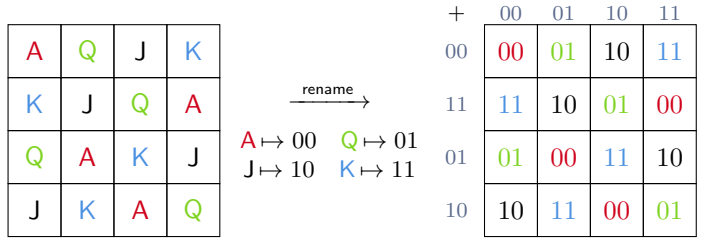


Figure: Rename the ranks and label the rows and columns by elements of $\mathbb{Z}_2 \times \mathbb{Z}_2$: every cell becomes row + column. No cell moves.

Which groups G have a Cayley table that decomposes into transversals?

For Groups, One Transversal Is Enough

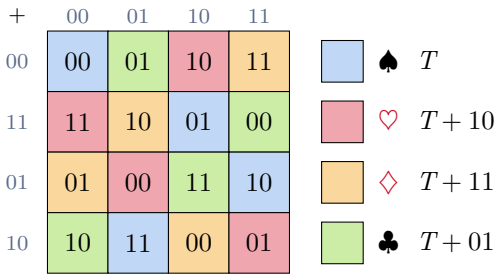


Figure: The four suits are the translates of one transversal T: shift every column label by the same element.

Fact 8

Let $T = \{(x, \phi(x)) : x \in G\}$ be a transversal of the Cayley table of G. For each $g \in G$,

$$T_g = \{(x, \phi(x)g) : x \in G\}$$

is again a transversal, and the n sets T_g partition the table.

The entries of T_g are those of T, multiplied on the right by g.

For a group, Euler's question asks for one rainbow perfect matching.

Complete Mappings: Why Mann Introduced Them

By 1942, the known pairs of orthogonal Latin squares — in the statistical tables of Fisher and Yates, and from Galois fields (Bose, 1938), (Stevens, 1939) — were all **based on groups**. Mann asked which groups can be used.

Definition 9 (Mann, 1942)

A bijection $\theta\colon G \to G$ is a **complete mapping** if $x \mapsto x\theta(x)$ is also a bijection.

In the Cayley table: in row x, take the cell in column $\theta(x)$. These n cells form a transversal, and every transversal arises this way.

Theorem 10 (Mann, 1942; Paige, 1951)

The Cayley table of G has an orthogonal mate $\iff$ G has a complete mapping θ. One mate is the Latin square $S(i,k) = ik\theta(k)$.

Proposition 2: Complete Mappings Are Rainbow Perfect Matchings

Let $K_{G,G}$ be the complete bipartite graph with two copies of G as sides, the edge (x, y) coloured xy. This is the Cayley table as a colouring.

Proposition 11

*A bijection $\theta : G \to G$ is a complete mapping $\iff$ the perfect matching $\{(x, \theta(x)) : x \in G\}$ of $K_{G,G}$ is **rainbow**. Every rainbow perfect matching arises in this way, from exactly one θ.*

Proof. A perfect matching of $K_{G,G}$ is the graph of a bijection θ; its colours are $x\theta(x)$, and they are distinct iff $x \mapsto x\theta(x)$ is injective, i.e. a bijection. □

Corollary 12

By the translates T_g (slide For Groups, One Transversal Is Enough) and Proposition 6: G has a complete mapping $\iff$ $K_{G,G}$ has a rainbow perfect matching $\iff$ $E(K_{G,G})$ splits into $|G|$ rainbow perfect matchings.

The First Results, 1942–1951

▷ (Mann, 1942) **No group of order** $4n+2$ has a complete mapping: groups can never refute Euler's conjecture.

▷ (Mann, 1942) New orthogonal pairs of order 12, from $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_3$ and from A_4, beyond the Galois-field method.

▷ (Paige, 1947) An abelian group has a complete mapping $\iff \sum_{x \in G} x = 0$.

▷ (Bateman, 1950) Every infinite group has one.

▷ (Paige, 1951) Every group of odd order has one: $\theta(x) = x$.

Theorem 13 (Paige, 1951)

If G has a complete mapping, then the product of all elements of G, in any order, lies in the commutator subgroup G'.

Paige conjectured the converse. He also conjectured that **no symmetric group** S_n has a complete mapping.

Two Groups of Order 4

$+$	00	01	10	11
00	00	01	10	11
11	11	10	01	00
01	01	00	11	10
10	10	11	00	01

$\mathbb{Z}_2 \times \mathbb{Z}_2$

$+$	0	1	2	3
0	0	1	2	3
1	1	2	3	0
2	2	3	0	1
3	3	0	1	2

$\mathbb{Z}_4$

Figure: Both tables are Latin squares of order 4. The first has a transversal (circled); the second has none.

$\mathbb{Z}_2 \times \mathbb{Z}_2$: $00 + 01 + 10 + 11 = 00$ Paige's condition holds

$\mathbb{Z}_4$: $0 + 1 + 2 + 3 = 2 \neq 0$ Paige's condition **fails**

Same order, different group structure.

Hall and Paige, 1955

Fact 14 (Hall–Paige, 1955)

The product of all elements of G lies in G' $\iff$ the Sylow 2-subgroups of G are trivial or non-cyclic.

For $\mathbb{Z}_4$ the Sylow 2-subgroup is $\mathbb{Z}_4$ itself: cyclic. For $\mathbb{Z}_2 \times \mathbb{Z}_2$ it is not.

▷ (Hall–Paige, 1955) every **solvable** group with this property has a complete mapping.

▷ (Hall–Paige, 1955) S_n $(n > 3)$ and A_n have complete mappings — **Paige's conjecture on S_n was false**.

Conjecture 3 (Hall–Paige, 1955)

A finite group G has a complete mapping $\iff$ its Sylow 2-subgroups are trivial or non-cyclic.

Proved in 2009, via the Classification

▷ (Hall–Paige, 1955) if a normal subgroup N and the quotient G/N both have complete mappings, so does G. This settles solvable groups.

▷ (Wilcox, 2009) a minimal counterexample is **simple**; groups of Lie type are handled uniformly. Only the Tits group and the 26 sporadic groups remain.

▷ (Evans, 2009) all of these except the fourth Janko group J_4, with computer algebra.

▷ (Bray, 2009; published 2020) J_4.

Theorem 15 (Wilcox, Evans, Bray, 2009)

The Hall–Paige conjecture is true.

A question about Latin squares, settled by the classification of finite simple groups.

A Proof by Counting

Theorem 16 (Eberhard–Manners–Mrazović, 2022)

If G has order n and satisfies the Hall–Paige condition, the number of complete mappings of G is $\big(e^{-1/2} + o(1)\big)\,|G^{ab}|\,\dfrac{n!^2}{n^n}$.

▷ $n! \cdot \dfrac{n!}{n^n}$ is the naive guess: $n!$ bijections ϕ, and $x \mapsto x\phi(x)$ is a bijection with "probability" $n!/n^n$.

▷ The count is positive for every large G: a proof of Hall–Paige with **no classification**, though not a construction.

▷ Combined with Wilcox's reduction, only the Mathieu groups M_{11} and M_{12} remain to be treated separately.

A Proof by Random Subsets

Keep only the rows in X, the columns in Y, and the symbols in Z of the Cayley table, where $|X| = |Y| = |Z|$. A **transversal** of what is left uses every symbol of Z once.

Fact 17

If it exists, then $\sum X + \sum Y = \sum Z$ in G^{ab} — the same sum argument as for Hall–Paige.

Theorem 18 (Müyesser–Pokrovskiy, 2025)

*Let G be large. If X, Y, Z are each **close to a random subset** of G (every element kept independently with probability p), then $\sum X + \sum Y = \sum Z$ in G^{ab} is also **sufficient**.*

$\triangleright$ $X = Y = Z = G$ ($p = 1$): the Hall–Paige conjecture for large G, with no classification.

The Dictionary

algebra / Latin squares	**graphs**
Latin square of order n	proper n-edge-colouring of $K_{n,n}$
transversal	rainbow perfect matching
orthogonal mate	decomposition into n rainbow perfect matchings
Cayley table of G	$K_{G,G}$, edge (x, y) coloured xy
complete mapping of G	rainbow perfect matching of $K_{G,G}$
number of complete mappings	number of rainbow perfect matchings
strong complete mapping	perfect matching rainbow in two colourings
transversal of a subsquare $A \times B$	rainbow perfect matching of $K_{A,B}$
orthomorphism with cycle type	rainbow cycle factor in $\mathrm{Cay}(G, G \setminus \{e\})$ (deck 02)

Beyond Hall–Paige: Three Questions

Problem 19 (How many?)

How many transversals does a Cayley table have, and which groups have the most?

Problem 20 (In which subsets?)

Delete some rows, columns and symbols. When does what is left still have a transversal?

Problem 21 (Of which shape?)

A complete mapping is a permutation of G. Which cycle types can it have?

Contents

1. From Euler to the Hall–Paige Conjecture

2. How Many Transversals? Semiqueens and Queens

3. Beyond Hall–Paige: Subsquares and Cycle Types

How Many Transversals?

Theorem 22 (Eberhard–Manners–Mrazović, 2019; Vardi's conjecture)

For odd n, the Cayley table of $\mathbb{Z}_n$ has $\left(e^{-1/2} + o(1)\right) n \dfrac{n!^2}{n^n}$ transversals.

Equivalently: the number of ways to place n non-attacking **semiqueens** on an $n \times n$ board wrapped into a torus. A semiqueen moves along rows, columns, and one family of diagonals.

▷ (Vardi, 1991) conjectured that the count lies between $c_1^n\, n!$ and $c_2^n\, n!$ for constants $0 < c_1 \leq c_2 < 1$; the lower bound was the hard part.

▷ (Eberhard–Manners–Mrazović, 2022) every group: the same constant $e^{-1/2}$, times $|G^{ab}|$.

Corollary 23 (Eberhard–Manners–Mrazović, 2022)

For large k, among all groups of order 2^k, the elementary abelian group $\mathbb{Z}_2^k$ has the most complete mappings.

The Eight-Queens Puzzle

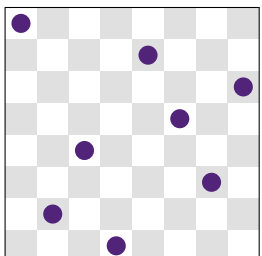


Figure: A solution.

Problem 24 (Bezzel, 1848)

*Place **eight queens** on a chessboard so that no two attack each other: no two share a row, a column, or a diagonal.*

▷ (Nauck, 1850) all 92 solutions; Gauss also worked on the puzzle.

▷ (Pauls, 1874) on an $n \times n$ board, n queens fit for every $n \notin \{2, 3\}$.

▷ Let $Q(n)$ be the number of solutions: $Q(8) = 92$. How does $Q(n)$ grow?

Queens as Permutations

Number the rows and the columns $1, \ldots, n$. With one queen in each row and each column, the queen of row j stands in column $\sigma(j)$ for a **permutation** σ. The square (j, c) lies on the diagonals $c + j = \text{const}$ and $c - j = \text{const}$. So:

Fact 25

σ is a solution $\iff$ the n numbers $\sigma(j) + j$ are distinct and the n numbers $\sigma(j) - j$ are distinct.

j	1	2	3	4	5	6	7	8
$\sigma(j)$	1	5	8	6	3	7	2	4
$\sigma(j) + j$	2	7	11	10	8	13	9	12
$\sigma(j) - j$	0	3	5	2	-2	1	-5	-4

The solution drawn on the previous slide, checked.

Pólya's Toroidal Board

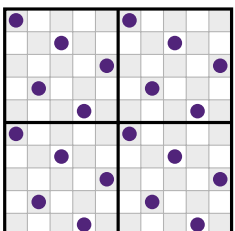


Figure: Five queens, repeated periodically: still no attacks.

Pólya (1918) studied the **doubly periodic** solutions: those that stay non-attacking when the board is repeated to fill the whole plane. Equivalently, the diagonals **wrap around**: the board is a torus, and

$$\sigma(j)+j \text{ and } \sigma(j)-j \text{ are distinct} \pmod{n}.$$

▷ The torus is the more regular problem: n rows, n columns and n diagonals each way.

▷ A toroidal solution is an ordinary one: $T(n) \le Q(n)$.

▷ (Pólya, 1918) $T(n) > 0 \iff n \equiv 1, 5 \pmod{6}$.

Semiqueens, Queens and Complete Mappings

on the $n \times n$ torus	**condition on σ (mod n)**	**in the group $\mathbb{Z}_n$**
n semiqueens	$\sigma(j) + j$ distinct	a **complete mapping**
n queens	$\sigma(j) + j$ and $\sigma(j) - j$ distinct	a **strong complete mapping**

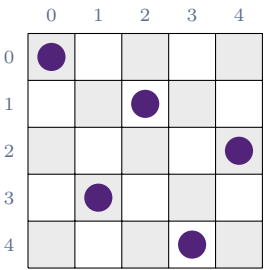


Figure: $\sigma(j) = 2j$ on $\mathbb{Z}_5$.

▷ Semiqueens exist $\iff$ n is odd: Hall–Paige for $\mathbb{Z}_n$.

▷ (Pólya, 1918) Queens exist $\iff$ $n \equiv 1, 5 \pmod 6$.

▷ In the figure $\sigma(j) + j = 3j$ and $\sigma(j) - j = j$ are both bijections of $\mathbb{Z}_5$.

Counting Queens

$Q(n)$: queens on the ordinary board. $T(n)$: queens on the torus.

- ▷ (Rivin–Vardi–Zimmermann, 1994) conjectured $\log Q(n)$, $\log T(n) = \Theta(n \log n)$.
- ▷ (Luria, 2017) $T(n) \le \big((1+o(1))\, ne^{-3}\big)^n$, conjectured to be tight.
- ▷ (Bowtell–Keevash, 2021) $T(n) = \big((1+o(1))\, ne^{-3}\big)^n$ for $n \equiv 1, 5 \pmod 6$.
- ▷ (Luria–Simkin, 2021), (Bowtell–Keevash, 2021) $Q(n) \ge \big((1-o(1))\, ne^{-3}\big)^n$.
- ▷ (Simkin, 2023) $Q(n) = \big((1 \pm o(1))\, ne^{-\alpha}\big)^n$ with $\alpha = 1.942 \pm 0.003$.

The tools are those of Hall–Paige: a random greedy process, then absorption.

Queens: Recent Progress

▷ (Simkin, 2023) the ordinary board: $Q(n) = \left((1 \pm o(1))\, ne^{-\alpha}\right)^n$ with $\alpha = 1.942 \pm 0.003$, about $(0.143\, n)^n$.

Theorem 26 (Glock–Munhá Correia–Sudakov, 2022)

For large n, any $n/60$ non-attacking queens can be completed to a solution; some $\approx n/4$ queens cannot.

The completion question goes back to Nauck (1850); the proof goes through **rainbow matchings** in bipartite graphs.

Theorem 27 (Dai–Kelly, 2024)

For all large n there is a reflecting n-queens configuration, answering questions of Klarner (1967) and Slater.

Open Problem 1: Strong Complete Mappings

A bijection $\phi\colon G \to G$ is a **strong complete mapping** if both $x \mapsto x\phi(x)$ and $x \mapsto x^{-1}\phi(x)$ are bijections. For $\mathbb{Z}_n$ these are the toroidal queens.

Evans

Which finite groups G have a bijection ϕ such that $x\phi(x)$ and $x^{-1}\phi(x)$ both run through all of G?

$\Updownarrow$ Proposition 11, applied to both colourings

Graph form

Colour each edge (x, y) of $K_{G,G}$ twice: by xy and by $x^{-1}y$. For which G is there a perfect matching that is **rainbow in both colourings**?

Known. $\mathbb{Z}_n$: exactly when $n \equiv 1, 5 \pmod 6$ (Pólya, 1918), via toroidal queens.

Contents

1. From Euler to the Hall–Paige Conjecture

2. How Many Transversals? Semiqueens and Queens

3. Beyond Hall–Paige: Subsquares and Cycle Types

Delete One Row and One Column

+	0	1	2	3
0	(0)	1	2	3
1	1	2	(3)	0
2	2	3	0	(1)
3	3	0	1	2

Figure: $\mathbb{Z}_4$: delete row 3, column 1 and symbol 2. The circled cells are a transversal of what is left.

A **near transversal** is a set of $n-1$ cells in distinct rows, distinct columns and with distinct symbols. $\mathbb{Z}_{2m}$ has no transversal, so $n-1$ is the best one can ask for.

Theorem 28 (Goddyn–Halasz, 2020)

Every Cayley table — indeed every Latin square main-class equivalent to one — has a near transversal.

This is the Ryser–Brualdi–Stein conjecture for Cayley tables.

In the language of random Hall–Paige: $X = G \setminus \{3\}$, $Y = G \setminus \{1\}$, $Z = G \setminus \{2\}$ and $\sum X + \sum Y = 3 + 1 = 0 = \sum Z$ in $\mathbb{Z}_4$.

▷ (Müyesser–Pokrovskiy, 2025) a second proof for large groups, this way.

Subsquares: Snevily's Conjecture

Keep the rows in A and the columns in B, with $|A| = |B| = k$: a $k \times k$ **subsquare** $A \times B$. A transversal of it is a bijection $\pi\colon A \to B$ with the k sums $a + \pi(a)$ distinct.

+	0	2	4
0	0	2	4
1	1	3	0
3	3	0	2

Figure: $\{0, 1, 3\} \times \{0, 2, 4\}$ in $\mathbb{Z}_5$: a transversal.

+	0	1	2	3
0	0	1	2	3
1	1	2	3	0
2	2	3	0	1
3	3	0	1	2

Figure: $\{0, 2\} \times \{0, 2\}$ in $\mathbb{Z}_4$: the table of $\mathbb{Z}_2$, none.

Deleting rows and columns should only make a transversal easier to find — **unless** what is left is itself the table of a group with no transversal, as on the right.

Snevily's Conjecture

Conjecture 4 (Snevily, 1999)

Let G be abelian and $A, B \subseteq G$ with $|A| = |B|$.

1. *If $|G|$ is odd, $A \times B$ has a transversal.*
2. *If $G \cong \mathbb{Z}_{2k}$, $A \times B$ has a transversal unless $g_1 + A = g_2 + B = H$ for some even-order cyclic subgroup H.*

▷ Part 1 says: in an abelian group of odd order, **every** subsquare has a transversal. For $k = |G|$ this is the odd case of Hall–Paige.

▷ Part 2 says: in $\mathbb{Z}_{2k}$ the only subsquares without one are translates of the table of an even-order cyclic subgroup.

Progress on Snevily's Conjecture

▷ (Alon, 2000) Part 1 for $\mathbb{Z}_p$ — even when A is a **sequence** of $k < p$ elements, with repetitions — by the Combinatorial Nullstellensatz.

▷ (Dasgupta–Károlyi–Serra–Szegedy, 2001) Part 1 for every cyclic group of odd order.

▷ (Arsovski, 2011) Part 1 for every abelian group of odd order, by linear algebra over characters.

▷ Part 2: no partial progress before 2025.

Part 2 cannot be stated for all even abelian groups:

Example 29 (Akbari–Alipour, 2004)

In $(\mathbb{Z}_2)^k$ take $a_1 \neq a_2$, $b_1 \neq b_2$ with $a_1 + a_2 + b_1 + b_2 = 0$, and $A = G \setminus \{a_1, a_2\}$, $B = G \setminus \{b_1, b_2\}$. Then $A \times B$ has no transversal — yet it is not the table of a subgroup.

All Groups, Large Subsquares

Theorem 30 (Müyesser–Pokrovskiy, 2025)

For n large, any group G and $A, B \subseteq G$ with $|A| = |B| = n$: $A \times B$ has a transversal unless, for some $g_1, g_2 \in G$ and subgroup $H \leq G$,

1. *H fails the Hall–Paige condition, $A = g_1 H$ and $B = H g_2$; or*
2. *$H \cong (\mathbb{Z}_2)^k$, $g_1 A = H \setminus \{a_1, a_2\}$, $g_2 B = H \setminus \{b_1, b_2\}$ with $a_1 + a_2 + b_1 + b_2 = 0$.*

▷ In case 1, $A \times B$ is a copy of the table of H, which has no transversal by Hall–Paige. Case 2 is Akbari–Alipour.

▷ **Nothing else** goes wrong. In particular, both parts of Snevily's conjecture hold for large subsquares.

Open Problem 2: Subsquares for Every n

Conjecture 5 (Müyesser–Pokrovskiy, Conj. 7.1)

*For **every** n: if $A, B \subseteq G$ with $|A| = |B| = n$, there is a bijection $\pi : A \to B$ with all $a\pi(a)$ distinct, unless for some $g_1, g_2 \in G$ and subgroup H: (1) H fails the Hall–Paige condition, $A = g_1 H$, $B = Hg_2$; or (2) $H \cong \mathbb{Z}_2^k$, $g_1 A = H \setminus \{a_1, a_2\}$, $g_2 B = H \setminus \{b_1, b_2\}$ with $a_1 + a_2 + b_1 + b_2 = 0$.*

$\Updownarrow$ Proposition 11, restricted to $A \times B$

Graph form

For every n: in $K_{G,G}$ with (x, y) coloured xy, the induced subgraph $K_{A,B}$ has a **rainbow perfect matching**, unless (1) or (2) holds.

Known. all large n (Müyesser–Pokrovskiy, 2025); both parts of Snevily's conjecture are special cases.

Open Problem 3: Hall–Paige Without the Classification

Conjecture 6 (Müyesser–Pokrovskiy, Conj. 7.2)

Let $|G| = n$ and $X, Y, Z \subseteq G$ with $|X| = |Y| = |Z| = n - 1$ and $\sum X + \sum Y = \sum Z$ in G^{ab}. Then there is a bijection $\phi : X \to Y$ with $\{x\phi(x) : x \in X\} = Z$.

⇕ Proposition 11; equivalently a perfect matching of the multiplication hypergraph

Graph form

In $K_{X,Y} \subseteq K_{G,G}$ (edge (x, y) coloured xy) there is a perfect matching using **each colour of Z exactly once**.

Known. true for large n when X, Y, Z are close to random sets, and for sets of size $n - O(n^{1-\varepsilon})$ in $\mathbb{Z}_n$ (Müyesser–Pokrovskiy). With $X = Y = Z = G \setminus \{e\}$ it gives Hall–Paige for every n without the classification.

Open Problem 4: How Much Can Be Deleted?

Müyesser–Pokrovskiy, Problem 7.4

Find the largest $g(n)$ such that: whenever $X, Y, Z \subseteq \mathbb{Z}_n$ have size $n - g(n)$ and $\sum X + \sum Y = \sum Z$, some bijection $\phi : X \to Y$ has $\{x + \phi(x)\} = Z$.



Proposition 11

Graph form

Largest $g(n)$ such that, for all such X, Y, Z, the graph $K_{X,Y}$ with (x, y) coloured $x + y$ has a perfect matching using each colour of Z once.

Known. $g(n) = \Omega(n)$ and $g(n) \leq n/2$ (Müyesser–Pokrovskiy, §7). "It would already be interesting to determine $g(n)$ asymptotically."

Open Problem 5: The Kézdy–Snevily Problem

Snevily, 1999; Kézdy–Snevily, 2002; MP, Problem 7.5

For $k < n$, any $a_1, \ldots, a_k \in \mathbb{Z}_n$ (not necessarily distinct) admit a permutation π with $a_{\pi(1)} + 1,\ a_{\pi(2)} + 2,\ \ldots,\ a_{\pi(k)} + k$ distinct.

$\Updownarrow$ a perfect matching $i \mapsto \pi(i)$ of $K_{k,k}$

Graph form

Colour the edge (i, j) of $K_{k,k}$ (both sides $\{1, \ldots, k\}$) by $a_i + j \in \mathbb{Z}_n$. For $k < n$, there is always a **rainbow perfect matching**.

The colouring is proper in j, but not in i when the a_i repeat — the new difficulty compared with Snevily's conjecture.

Known. n prime (Alon, 2000); $k = n - 1$ (Hall, 1952); $2k \leq n + 1$ (Kézdy–Snevily); large k without repetitions (MP). "Wide open" otherwise.

Of Which Shape? Orthomorphisms and Cycle Types

Definition 31

A bijection $\psi\colon G \to G$ is an **orthomorphism** if $x \mapsto x^{-1}\psi(x)$ is also a bijection. Equivalently: in row x of the Cayley table, take the cell with symbol $\psi(x)$; these cells form a transversal (Wanless).

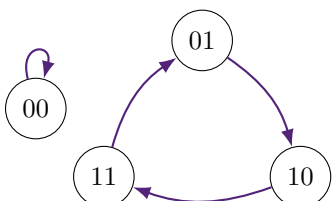


Figure: The ♠ transversal as a permutation of $\mathbb{Z}_2 \times \mathbb{Z}_2$.

An orthomorphism is a **permutation**, so it has a cycle type. Why ask for a prescribed one?

▷ One cycle through all non-identity elements: Ringel's solution of the Heawood map-colouring problem (Müyesser).

▷ Disjoint 6-cycles: cyclic Steiner triple systems (Müyesser).

The Friedlander–Gordon–Tannenbaum Conjecture

Conjecture 7 (Friedlander–Gordon–Tannenbaum, 1981)

Let G be abelian of order n satisfying the Hall–Paige condition, and $k \geq 2$ with $k \mid n-1$. Then some orthomorphism of G fixes 0 and permutes the other elements as disjoint k-cycles.

Our example is the case $G = \mathbb{Z}_2 \times \mathbb{Z}_2$, $k = 3$.

▷ (Friedlander–Gordon–Tannenbaum, 1981) $|G| \leq 15$, and abelian p-groups with $p \geq 3$.

▷ Even $k = 3$ with G cyclic was open for forty years.

Theorem 32 (Müyesser, 2023)

The conjecture holds for all sufficiently large groups.

Proposition 3: Cycle Types Are Rainbow Cycle Factors

Recall deck 02: $\mathrm{Cay}(G, G \setminus \{e\})$ has the arc $x \to xs$ of colour s; equivalently the arc $x \to y$ has colour $x^{-1}y$.

Proposition 33

*An orthomorphism ψ of G with $\psi(e) = e$ that permutes $G \setminus \{e\}$ in cycles of lengths $k_1, k_2, \ldots$ is the same as a spanning family of vertex-disjoint directed cycles of lengths $k_1, k_2, \ldots$ on $G \setminus \{e\}$ in $\mathrm{Cay}(G, G \setminus \{e\})$ that is **rainbow**.*

Proof. Given ψ, use the arcs $x \to \psi(x)$, $x \neq e$; their colours $x^{-1}\psi(x)$ are distinct (orthomorphism) and $\neq e$ (since $e^{-1}\psi(e) = e$). Conversely, a rainbow cycle factor on $G \setminus \{e\}$ uses $|G| - 1$ distinct non-identity colours, i.e. each once; set $\psi(x) =$ the successor of x and $\psi(e) = e$: then $x \mapsto x^{-1}\psi(x)$ takes every value once. □

One cycle of length $|G| - 1$: Ringel's R-sequencings (deck 02). All 3-cycles: the case $k = 3$ of the FGT conjecture.

Open Problem 6: FGT for All Groups

Conjecture 8 (Müyesser, Conj. 7.1)

Let G be a sufficiently large group satisfying the Hall–Paige condition, and $k \geq 3$ with $k \mid n-1$. Then some orthomorphism of G fixes e and permutes the other elements in disjoint k-cycles.

⇕ Proposition 33

Graph form

Under the same hypotheses, $\mathrm{Cay}(G, G \setminus \{e\})$ contains a **rainbow directed C_k-factor** on the vertex set $G \setminus \{e\}$.

Known. abelian groups, all large (Müyesser, 2023); the proof's high-girth part does not use commutativity (§7.1). Small abelian cases: $|G| \leq 15$ and p-groups with $p \geq 3$ (FGT, 1981).

Open Problems 7 and 8: Beyond Groups, Beyond 3-Cycles

Conjecture 9 (Müyesser, Conj. 7.3)

Every complete graph K_n, properly coloured with n colours, contains a ***rainbow*** *family of disjoint triangles covering all but at most C vertices, for an absolute constant C.*

Known. with C replaced by $n^{1-\varepsilon}$, by the Rödl nibble; no example rules out $C = 2$ (§7.1). It is the case $k = 3$ of FGT with Cayley tables replaced by all proper colourings.

Müyesser, §7.2

Which cycle types can an orthomorphism of an abelian group have? Equivalently (Proposition 33): which families of directed cycle lengths occur in **rainbow cycle factors** of $\mathrm{Cay}(G, G \setminus \{e\})$ on $G \setminus \{e\}$?

Known. many mixed types with each cycle length used linearly often (§7.2).

Open Problem 9: One Equation, Many Problems

Fix an integer matrix M ($r \times m$) and an abelian group G of order n.

Müyesser, §7.3

For which M, G are there n solutions $v \in G^m$ of $Mv = 0$ such that in every coordinate each element of G appears exactly once? Open already for entries in $\{-1, 0, 1\}$.

$\Updownarrow$ solutions = hyperedges

Hypergraph form

Let $\mathcal{H}_M$ be the m-partite m-uniform hypergraph with parts $G \times \{1\}, \ldots, G \times \{m\}$ and an edge $\{(v_1, 1), \ldots, (v_m, m)\}$ for each solution v. When does $\mathcal{H}_M$ have a **perfect matching**?

Hall–Paige: $\begin{pmatrix} 1 & -1 & -1 \end{pmatrix}$ toroidal queens: $\begin{pmatrix} 1 & 1 & -1 & 0 \\ 1 & -1 & 0 & -1 \end{pmatrix}$ FGT, $k = 3$: a 3×6 matrix

The hypergraph form is Müyesser's own ("we can equivalently phrase this as a hypergraph matching problem", §7.3).

Summary of Open Problems

How many?

▷ Strong complete mappings: which groups? (Evans)

In which subsets?

▷ The subsquare characterisation for every n.

▷ Hall–Paige for every n without the classification.

▷ Kézdy–Snevily.

Of which shape?

▷ FGT for non-abelian groups.

▷ Rainbow triangle factors in proper colourings.

▷ All cycle types.

For large groups the algebraic obstruction is the only one. For small groups, and beyond groups, we do not know.

The People Behind the Recent Results



Benny Sudakov
Professor, ETH Zürich
PhD Tel Aviv 1999, advisor Noga Alon
n-queens completion (2022)



Peter Keevash
Professor, University of Oxford (Mansfield College)
PhD Princeton 2004, advisor Benny Sudakov
Toroidal n-queens (2021); existence of designs



Candida Bowtell
Assistant Professor, University of Birmingham
DPhil Oxford 2022, advisor Peter Keevash; postdoc at Warwick with R. Montgomery
Toroidal n-queens (2021)



Alexey Pokrovskiy
Associate Professor, University College London
PhD LSE 2013, advisors J. van den Heuvel and J. Skokan; postdocs FU Berlin, ETH
Random Hall–Paige (2025); European Prize in Combinatorics 2019



Alp Müyesser
Titchmarsh Research Fellow, University of Oxford; JRF, New College
PhD UCL 2024, advisor Alexey Pokrovskiy
Random Hall–Paige (2025); the FGT conjecture (2023)



Richard Montgomery
Professor, University of Warwick
PhD Cambridge 2015, advisor Andrew Thomason
Ryser–Brualdi–Stein for large even n (2023); EMS Prize 2024

📖 Positions as listed on their institutional pages and CVs; doctorates from the Mathematics Genealogy Project (2026).

Doctoral Lineage

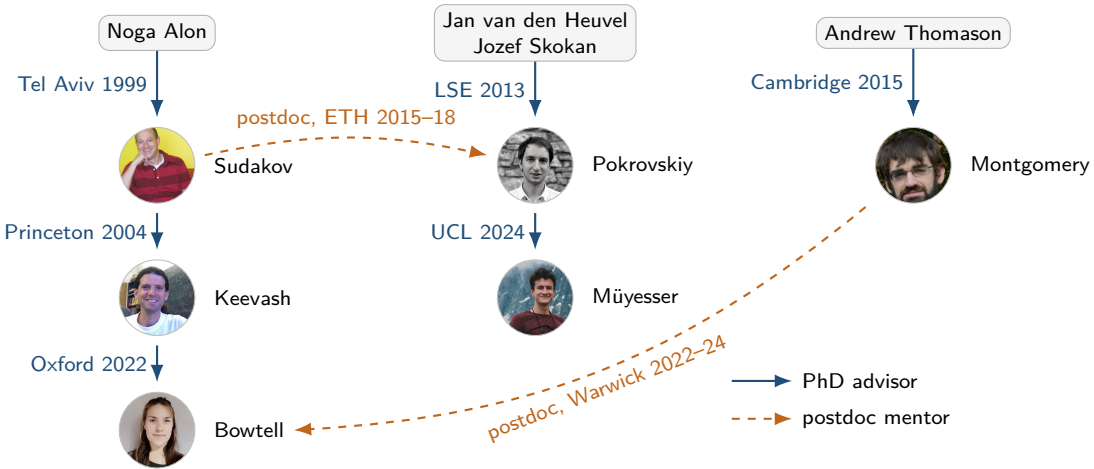


Figure: Doctoral advisors (solid) and postdoctoral mentors (dashed).

Thanks !

References I

Cayley tables and complete mappings

R. A. Bailey. *Latin squares*. G. C. Steward lecture, Gonville and Caius College, Cambridge, 7 March 2013 (slides).

A. Dey. *The problem of 36 officers, Euler's conjecture and related matters*. Special Proceedings of the 19th Annual Conference of SSCA (2017), 1–7.

J. Ozanam. *Récréations mathématiques et physiques*, vol. 1, 1723.

L. Euler. *Recherches sur une nouvelle espèce de quarrés magiques*. Verh. Zeeuwsch. Genootsch. Wetensch. Vlissingen 9 (1782), 85–239.

G. Tarry. *Le problème des 36 officiers*. Secrétariat de l'Association française pour l'avancement des sciences, 1900.

R. C. Bose, S. S. Shrikhande. *On the falsity of Euler's conjecture about the non-existence of two orthogonal Latin squares of order* $4t + 2$. Proc. Natl. Acad. Sci. USA 45 (1959), 734–737.

R. C. Bose, S. S. Shrikhande, E. T. Parker. *Further results on the construction of mutually orthogonal Latin squares and the falsity of Euler's conjecture*. Canad. J. Math. 12 (1960), 189–203.

I. M. Wanless. *Transversals in Latin squares*. Quasigroups Related Systems 15 (2007), 169–190. arXiv:0903.5142

H. J. Ryser. *Neuere Probleme der Kombinatorik*. Vorträge über Kombinatorik, Oberwolfach (1967), 69–91.

References II

S. K. Stein. *Transversals of Latin squares and their generalizations*. Pacific J. Math. 59 (1975), 567–575.

R. Montgomery. *A proof of the Ryser–Brualdi–Stein conjecture for large even n*. arXiv:2310.19779

R. A. Fisher, F. Yates. *Statistical Tables for Agricultural, Biological, and Medical Research*. Oliver and Boyd, Edinburgh.

R. C. Bose. *On the application of the properties of Galois fields to the problem of construction of hyper-Graeco-Latin squares*. Sankhyā (1938).

W. L. Stevens. *The completely orthogonalized Latin square*. Annals of Eugenics (1939).

H. B. Mann. *The construction of orthogonal Latin squares*. Ann. Math. Statistics 13 (1942), 418–423.

H. B. Mann. *On orthogonal Latin squares*. Bull. Amer. Math. Soc. 50 (1944), 249–257.

L. J. Paige. *A note on finite abelian groups*. Bull. Amer. Math. Soc. 53 (1947), 590–593.

P. Bateman. *Complete mappings of infinite groups*. Amer. Math. Monthly 57 (1950), 621–622.

L. J. Paige. *Complete mappings of finite groups*. Pacific J. Math. 1 (1951), 111–116.

R. Montgomery. *Transversals in Latin squares*. In: Surveys in Combinatorics 2024, London Math. Soc. Lecture Note Ser., Cambridge Univ. Press, 131–158. arXiv:2406.19873

References III

M. Hall. *A combinatorial problem on abelian groups*. Proc. Amer. Math. Soc. 3 (1952), 584–587.

M. Hall, L. J. Paige. *Complete mappings of finite groups*. Pacific J. Math. 5 (1955), 541–549.

S. Wilcox. *Reduction of the Hall–Paige conjecture to sporadic simple groups*. J. Algebra 321 (2009), 1407–1428. arXiv:1010.1323

A. B. Evans. *The admissibility of sporadic simple groups*. J. Algebra 321 (2009), 105–116.

J. N. Bray, Q. Cai, P. J. Cameron, P. Spiga, H. Zhang. *The Hall–Paige conjecture, and synchronization for affine and diagonal groups*. J. Algebra 545 (2020), 27–42. arXiv:1811.12671

S. Eberhard, F. Manners, R. Mrazović. *An asymptotic for the Hall–Paige conjecture*. Adv. Math. 404 (2022), 108423. arXiv:2003.01798

A. Müyesser, A. Pokrovskiy. *A random Hall–Paige conjecture*. Invent. Math. 240 (2025), 779–867. arXiv:2204.09666

L. Goddyn, K. Halasz. *All group-based Latin squares possess near transversals*. J. Combin. Des. 28 (2020), 358–365. arXiv:1905.07071

H. S. Snevily. *The Cayley addition table of* $\mathbb{Z}_n$. Amer. Math. Monthly 106 (1999), 584–585.

References IV

N. Alon. *Additive Latin transversals*. Israel J. Math. 117 (2000), 125–130.

S. Dasgupta, G. Károlyi, O. Serra, B. Szegedy. *Transversals of additive Latin squares*. Israel J. Math. 126 (2001), 17–28.

B. Arsovski. *A proof of Snevily's conjecture*. Israel J. Math. 182 (2011), 505–508.

S. Akbari, A. Alipour. *Transversals and multicolored matchings*. J. Combin. Des. 12 (2004), 325–332.

I. Vardi. *Computational Recreations in Mathematica*. Addison-Wesley, 1991.

S. Eberhard, F. Manners, R. Mrazović. *Additive triples of bijections, or the toroidal semiqueens problem*. J. Eur. Math. Soc. (2019), 441–463. arXiv:1510.05987

G. Pólya. *Über die "doppelt-periodischen" Lösungen des n-Damen-Problems*. In: W. Ahrens (ed.), Mathematische Unterhaltungen und Spiele 2, Teubner, 1918, 364–374.

M. Bezzel ("Schachfreund"). Proposal of the eight-queens problem. Berliner Schachzeitung 3 (1848), 363.

F. Nauck. *Briefwechseln mit allen für alle*. Illustrirte Zeitung 15 (1850), 182.

E. Pauls. *Das Maximalproblem der Damen auf dem Schachbrete*. Deutsche Schachzeitung 29 (1874), 129–134, 257–267.

References V

I. Rivin, I. Vardi, P. Zimmermann. *The n-queens problem*. Amer. Math. Monthly 101 (1994), 629–639.

Z. Luria. *New bounds on the number of n-queens configurations*. arXiv:1705.05225

Z. Luria, M. Simkin. *A lower bound for the n-queens problem*. arXiv:2105.11431

M. Simkin. *The number of n-queens configurations*. Adv. Math. (2023). arXiv:2107.13460

S. Glock, D. Munhá Correia, B. Sudakov. *The n-queens completion problem*. arXiv:2111.11402

T. Dai, T. Kelly. *On the existence of reflecting n-queens configurations*. Forum Math. Sigma (2024). arXiv:2407.12742

A. B. Evans. *The existence of strong complete mappings of finite groups: a survey*. Discrete Math. 313 (2013), 1191–1196.

R. J. Friedlander, B. Gordon, P. Tannenbaum. *Partitions of groups and complete mappings*. Pacific J. Math. 92 (1981), 283–293.

A. Müyesser. *Cycle type in Hall–Paige: a proof of the Friedlander–Gordon–Tannenbaum conjecture*. Forum Math. Sigma 14 (2026), e50. arXiv:2303.16157

References VI

A. E. Kézdy, H. S. Snevily. *Distinct sums modulo n and tree embeddings*. Combin. Probab. Comput. 11 (2002), 35–42.

C. Bowtell, P. Keevash. *The n-queens problem*. arXiv:2109.08083